

MOFA: Exploring Asymmetric Mixed Reality Design Strategy for Co-located Multiplayer Between Handheld and Head-mounted Augmented Reality

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ABSTRACT

For co-located multiplayer asymmetric Mixed Reality (MR) scenarios involving both Handheld Augmented Reality (HAR) and head-mounted Stereoscopic Augmented Reality (StAR) devices, we propose a design strategy to distinguish the roles of different players based on the affordances, limitations, and capabilities of these two kinds of AR devices. Specifically, the roles include: a HAR player as a third-person passive participant (termed spectator), a third-person active participant (termed puppeteer), a first-person participant, and a first-person partial-information participant; and a StAR player as a first-person participant and a first-person partial-information participant. In order to explore this concept, we designed four multiplayer MR game prototypes: The Duck, The Ghost, The Dragon, and The Duel to demonstrate strategies for creating interesting, engaging, and strategic cooperative experiences between players.

CCS CONCEPTS

• **Human-centered computing** → Interaction design theory, concepts and paradigms; **Mixed / augmented reality**.

KEYWORDS

Augmented reality, mixed reality, collocated multiplayer, asymmetric game design

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1 INTRODUCTION AND RELATED WORKS

Handheld Augmented Reality (HAR) devices, such as those available on ubiquitous iOS (ARKit) and Android (ARCore) smartphones, are widely accessible. In contrast, head-mounted Stereoscopic Augmented Reality (StAR) devices, such as HoloLens and Magic Leap, are still relatively inaccessible in the consumer market. In the near future, it is likely that co-located multiplayer scenarios involving both StAR and HAR devices simultaneously will become more common.

Early research has explored both co-located HAR multiplayer [1] and co-located StAR multiplayer [5]. Additionally, research has focused on the use of HAR as a cross-device interactive controller for StAR devices in singleplayer settings, as demonstrated in [8], [7], and [3]. However, there has been little research on the mixture of HAR and StAR devices for asymmetric co-located multiplayer.

This study presents two major contributions: (1) a design strategy is proposed to distinguish the roles of different players based on the affordances, limitations, and capabilities of StAR and HAR devices, and (2) four multiplayer MR game prototypes were designed to demonstrate the proposed design strategy, thus demonstrating the potential of leveraging the physical affordance of the AR devices to create asymmetric, interesting, engaging, and strategic cooperative experiences between players.

2 DESIGN STRATEGIES AND PROTOTYPES

The majority of traditional video games present their character to the player through either a first-person or third-person camera view. A first-person game is one in which the player experiences the game from the viewpoint of the character they are controlling, while in a third-person game, the player views their character as an outside observer, with an auto-following camera typically situated over-the-shoulder or behind-the-back of their character.

StAR devices offer a unique physical affordance for providing an immersive experience to players through the utilization of head-tracking and motion tracking to accurately track the position and orientation of the player's head in real-time. This data is then used to generate a first-person view (FPV) that corresponds to the concept of egocentric view or internal piloting. However, controlling a character in third-person view (TPV), which corresponds to allocentric view or external piloting, is more difficult due to inconsistencies

between the viewpoints of the character and the player. As a result, StAR devices provide physical affordances for the egocentric view of the game, but not for the allocentric view. [6].

HAR devices offer physical affordances that support both FPV and TPV gaming experiences. HAR devices allow players to observe the physical world un-augmented, while also providing the capability to observe the augmented scene on the screen. The device is held by the player, thus separating the device's movement from the player's head movement. This provides the affordance of a handheld monitor, wherein the player can either pretend to be the main character or observe the character from a third-person perspective. As such, both FPV and TPV for HAR devices are potentially possible [4] [2].

Our methodology is based on the difference of the affordance of HAR and StAR devices to assign asymmetric roles for players. We proposed 2 possible design patterns for the roles of StAR player

- as a first-person participant
- as a first-person partial-information participant

also we proposed 4 possible design patterns for the roles of HAR player

- as a third-person passive participant (termed the spectator)
- as a third-person active participant (termed the puppeteer)
- as a first-person participant
- as a first-person partial-information participant

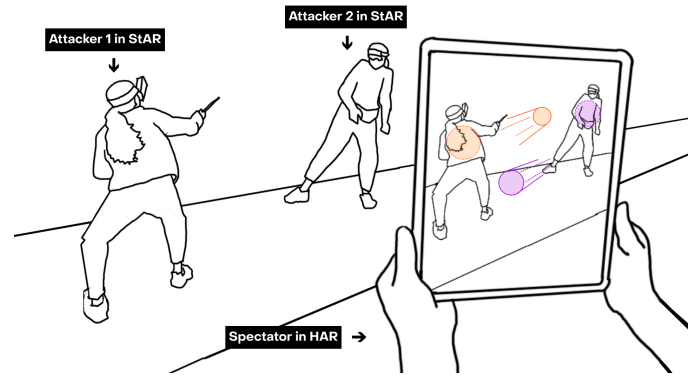
We conducted a study to investigate the effects of assigning differing roles, perspectives, and levels of information control in a co-located asymmetric mixed reality game. Four distinct games were designed to explore and demonstrate the game patterns impacted by these variables.

2.1 Demo 1: MOFA - The Duel

Drawing inspiration from the Wizardry Duel in the Harry Potter series and dodgeball, we developed a game called MOFA – The Duel. The premise of the game is that players compete against each other by throwing virtual magic balls at one another while taking care to avoid being hit themselves. As it is not possible to cause physical harm to a real player within the StAR environment, we created virtual shields in front of each player that can be destroyed by the magic balls. The goal of the game is to destroy as many shields as possible without allowing one's own shield to be destroyed within time limit of 90 seconds.

HAR players are spectators who can use handheld devices such as cameras and recorders to document the battle. The information required to play the game is asymmetrically distributed, with each player only knowing their own current parameters such as magic spell parameters and cooling down time, while spectators should be informed of all the information concerning the game. It is essential that spectators are provided with full information about the current situation to enable them to understand the situation adequately.

The two main players as first-person partial-information participants should use StAR devices in a way that allows them to act as themselves, acting as wizards. The immersive use of StAR devices in FPV will allow players to view the battle from their own perspective. The spectator as a third-person passive participant, on the other hand, will not be involved in the battle, but can observe the battle from any perspective he or she wants in third person



(a) Spectator views the battle

PARTICIPANT	ROLE	VIEW	DEVICE	INFORMATION	ACTION
Player 1	Attacker	First-Person	StAR	Partial	Attack
Player 2	Attacker	First-Person	StAR	Partial	Attack
Player 3	Spectator	Third-Person	HAR	Full	View

(b) Roles

Figure 1: MOFA - The Duel

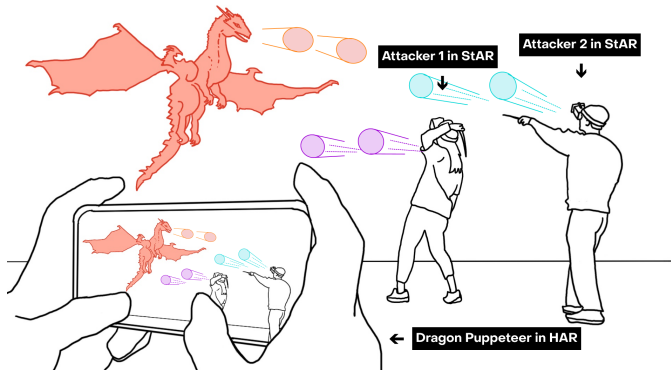
view. Additionally, the spectator can act as a cameraman, recording the battle as it takes place. Therefore, the spectator will use the HAR device in a natural way.

2.2 Demo 2: MOFA - The Dragon

In this demonstration, we explore the design strategy of an active third-person perspective for the HAR player. We refer to this HAR player as the puppeteer and through the use of the handheld device, the HAR player are able to control agents such as dragons. In an effort to captivate users, we decided to incorporate the classic dragon-hunting myth into our game.

We chose to use handheld devices for the puppeteer because they provide players with an expansive field of vision, efficient input methods and support for intricate interactions. This allows the players to operate their agents with ease, without the inefficiencies or inaccuracy of hand gesture control. The third-person view of puppeteers offers them perspective freedom, enabling them to have abilities that go beyond the constraints of the physical body, such as flying. This unique set-up increases the strength of the agent and puts the players in a situation where they are pitted against opponents that are far more powerful than themselves.

Furthermore, by using StAR devices, players are granted an intensely immersive experience that allows them to indulge in the dragon-hunting fantasy with a realistic perspective and heightened sense of spatial awareness. For example, in our game MOFA - The Dragon, players possess the power to launch airborne items and the dragon is able to soar and race through the air at superhuman speed.



(a) Attacker fights against Dragon Puppeteer

PARTICIPANT	ROLE	VIEW	DEVICE	INFORMATION	ACTION
Player 1	Attacker	First-Person	StAR	Full	Attack
Player 2	Attacker	First-Person	StAR	Full	Attack
Player 3	Dragon Puppeteer	Third-Person	HAR	Full	Maneuver & Attack

(b) Roles

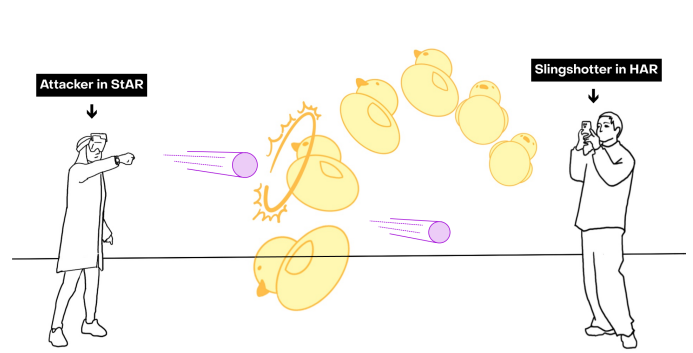
Figure 2: MOFA - The Dragon

During our user tests, we observed that players employed various tactics to win, such as dispersing their position, actively running and luring the enemy, which encourages cooperation among the players.

2.3 Demo 3: MOFA - The Duck

This demonstration seeks to explore the potential of first-person perspective for the HAR player. It is inspired by the classic Nintendo game Duck Hunt, as well as the sport of waterfowl hunting. The game is a two-player combat, in which the StAR player acts as the attacker and the HAR player as the slingshotter. The goal of the slingshotter is to throw the virtual duck into the air in such a way that it follows a parabolic trajectory, without being hit by the attacker's magic ball. On the other hand, the goal of the attacker is to predict the trajectory of the duck and throw the magic ball to hit it in the air.

In terms of interaction, the HAR Player interacts with the game by sliding their finger up on the touchscreen, thus providing the affordance of a functioning slingshot. Handheld devices provide greater flexibility in movement, tracking hand motions to throw the duck in any position they want. Furthermore, these HAR devices provide tactile feedback which drastically increases the accuracy and affordance of interactions that cannot be achieved with head-mounted devices. Head-mounted StAR devices, on the other hand, depend upon the user's physical body to dictate their role. However, they provide a more immersive spatial vision, thus allowing them



(a) Attacker aims at the Duck

PARTICIPANT	ROLE	VIEW	DEVICE	INFORMATION	ACTION
Player 1	Slingshotter	First-Person	HAR	Full	Throw
Player 2	Attacker	First-Person	StAR	Full	Attack

(b) Roles

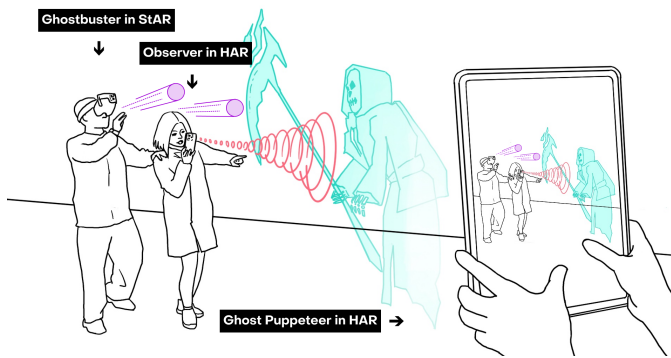
Figure 3: MOFA - The Duck

to make more accurate judgments on the duck's direction and hit the target.

We have taken into consideration the natural and predetermined asymmetric differences between handheld and head-mounted devices in terms of usage posture, interaction mode, and field of view that users receive, and adjusted the game accordingly. This creates cognitive-friendly interactive modes that enable players using both handheld and head-mounted devices to compete fairly under the regulation of game rules.

2.4 Demo 4: MOFA - The Ghost

This demonstration was inspired by the iconic film Ghostbusters. In this game, three players were situated in order to explore a design strategy for partial information, which could increase the collaboration between co-located multiplayer by taking advantage of the capabilities of their various devices. The players included a StAR player, referred to as the 'ghostbuster', and a HAR player, referred to as the 'detector', who were tasked with working together to battle an invisible adversary. This adversary, referred to as the 'ghost', was controlled by another HAR player referred to as the 'ghost puppeteer'. By default, the ghost was invisible to both players, with the HAR player serving as the 'detector', who was able to fire a detection wave in order to make the ghost visible, while the person wearing the StAR device functioned as the 'ghostbuster', who could throw magic call in order to harm the ghost. However, the 'ghostbuster' could not see the ghost until the 'detector' had managed to detect it for a few seconds. In order to completely



(a) Detector is finding the Ghost

PARTICIPANT	ROLE	VIEW	DEVICE	INFORMATION	ACTION
Player 1	Observer	First-Person	HAR	Partial	Detect
Player 2	Ghostbuster	First-Person	StAR	Partial	Attack
Player 3	Ghost Puppeteer	Third-Person	HAR	Full	Maneuver & Attack

(b) Roles

Figure 4: MOFA - The Ghost

defeat the enemy, players with different devices had to find ways to cooperate swiftly before being eliminated.

During user testing, it was observed that most players chose to get close in physical space in order to quickly exchange visual data, such as the position of the opponent. Simultaneously, players were seen to form an emotional dependence on each other. During the search for enemies, the ‘ghostbuster’ would usually stand behind or off to the side of the ‘detector’, appearing more nervous. When attacking, the ‘detector’ would often rely on the ‘attacker’ being close by. It was discovered that the asymmetry between the two first-person participants facilitated cooperation between them.

3 TECHNICAL REALIZATION

To create the prototype game, we employed the Unity3D game engine to develop apps that are compatible with iOS devices. For our StAR devices, we utilized multiple HoloKits in conjunction with the iPhone 14 Pro’s LiDAR Scanner and Apple Watch as gesture controller. The HoloKit X is an iPhone accessory that enables the phone to be transformed into an optical see-through StAR headset. For our HAR devices, we utilized the iPhone Pro series with the LiDAR Scanner.

In order to create a local network connection between devices, we utilized Apple’s MultipeerConnectivity technology. This technology enables devices to exchange data through both Bluetooth and Wi-Fi without the need for an internet connection, thus allowing nearby devices to remain connected to each other.

Our proposed technique, Spatial Anchor Sharing, enables the facilitation of a co-located multiplayer augmented reality experience with relative ease. This is achieved through the implementation of a host-client architecture local network, the calculation of the timestamp offset of all connected client devices relative to the host device, and the synchronization of the coordinate systems of all nearby devices in order to ensure that all devices display the same virtual content at the same physical location. This process is further expedited by the inclusion of a QR code, allowing for the instant establishment of a shared mixed reality experience without the requirement of an internet connection or pre-scanning of spatial maps.

4 DISCUSSION AND CONCLUSION

In this paper, we propose a design strategy that distinguishes the roles of different players in relation to their type of AR device. This strategy leads to the creation of four games that utilize people’s familiarity with the physical affordances of the AR device to designate digital roles. We contend that this approach presents a wealth of exciting potential opportunities for future designers of Multiplayer Mixed Reality (MR) games.

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